American International University-Bangladesh (AIUB)
Faculty of Engineering
Department of CSE, EEE, and CoE

EEE4103 MICROPROCESSOR AND EMBEDDED SYSTEM
COURSE CAPSTONE PROJECT PROPOSAL FORM

SEMESTER: SPRING 2025-26

PROJECT TITLE: (must be an embedded system designed using Arduino/STM32/Raspberry Pi with other necessary sensors, actuators, components, etc. Both hardware implementation and simulation must be shown by the end of the semester. However, the proposal form should be submitted within the next two weeks from now on) [2 MARKS]

SURVEY: (to develop a process for complex engineering problems considering cultural and societal factors (use **Google or Microsoft Forms** to design questions, show the outcomes in the form of a pie/bar chart, show the links of questions, **minimum 10 questions, and 40 respondents**), present survey link [7 MARKS]

AIMS AND OBJECTIVES OF THE PROJECT: (there must be a minimum of three aims and objectives of your work using measurable **action verbs** of Bloom's Taxonomy, for example, design, simulate, implement, analyze, etc.) [4 MARKS]

LITERATURE REVIEW: (must be related to embedded system design with other necessary citations from the most recently published papers. At least 5 most recent publications must be collected, studied, identified the problems, and then discussed in this section) [5 MARKS]

EXPERIMENTAL BLOCK DIAGRAM: (demonstrate how this project is interconnected with various components and how various signals flow among them using text boxes and arrows) [5 MARKS]

POSSIBLE OUTCOMES OF THE PROJECT: (describe how this project can help society and improve the culture of society) [4 MARKS]

PROJECT TIMELINE (GANTT CHART): describe how this project will progress with time by breaking the whole task into several components concerning some specific time using a horizontal or vertical bar graph) [5 MARKS]

REFERENCES: (only published paper-based references are allowed with the IEEE format, **don't use YouTube links, Wikipedia, or any random websites for references**; however, datasheets, databases, can be used as references with the appropriate formats, links, and access dates): [3 MARKS]

Sample for the conference paper:

[1] M. H. Bhuyan and Q. D. M. Khosru, "Linear Asymmetric Pocket Profile Based Pinch Off Voltage Model for Nano Scale n-MOSFET," Proceedings of the IEEE sponsored International Conference on Electrical, Computer and Communication Engineering (ICECCE2017), organized by the Chittagong University of Engineering and Technology (CUET), Cox's Bazar, Bangladesh, 16-18 February 2017, pp. 28-32.

Sample for the journal paper:

[1] M. H. Bhuyan and Q. D. M. Khosru, "Effects of Temperature on Reverse Short Channel Effect in Pocket Implanted Sub-100 nm n-MOSFET," Journal of Materials Science and Engineering, USA, 1934-8959, vol. 4, no 7, July 2010, pp. 18-23, DOI:10.17265/2161-6213/2010.07.004.

Instructions:

1. There is **no definite format for writing the course project proposal**, but students must follow the instructions mentioned above properly.
2. Fill in the form accurately with all necessary information.
3. Make a color print on this form.
4. Figures, tables, charts, circuit diagrams, block diagrams, and wave shapes must be color printed.
5. The survey form links with the answers that must be provided in the proposal form.

FOR FACULTY USE ONLY

COMMENTS BY COURSE TEACHER:

COURSE TEACHER'S NAME **COURSE TEACHER'S SIGNATURE** **DATE**

GROUP MEMBERS

(Maximum 6 students are permitted to carry out a single Project. However, depending on the capability of the students, 4 students may be allowed, but not fewer than that.

NAME: **Md. Shahariar Shiam** **ID #: 23-50496-1** **PROGRAM:** EEE/CoE/CSE **EMAIL: 23-50496-1@student.aiub.edu**	**NAME**: **Md. Shakib All Hassan** **ID #: 23-51407-1** **PROGRAM:** EEE/CoE/CSE **EMAIL: 23-51407-1@student.aiub.edu**
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Assessment Materials and Marks Allocation:

COs	Assessment Materials	POIs	Marks
CO3	Course Capstone Proposal Form	P.c.2.C6	35

Semester:	Spring 2025-26	**Sec:**	O
Faculty Member:	**Dr. Md. Rukonuzzaman**		

Capstone Project Title:		**An Arduino Based Dual Mode Omni Directional Rover for Intelligent Navigation and Obstacle Avoidance**	
Project Group No.		**8**	
Sl #	**Student ID #**	**Student Name**	**Obtained Marks**
1.	**23-50496-1**	**MD. SHAHARIAR SHIAM**	
2.	**23-51407-1**	**MD. SAKIB ALL HASAN**	
3.	**23-54596-3**	**MD. AHIQUR RAHMAN**	
4.	**23-53572-3**	**ADIBA AKRAM**	

Assessment Rubrics:

KPIs	Excellent [2]	Proficient [1.5]	Good [1]	Acceptable [0.5]	Unacceptable [0]	No Response [0]	Secured Marks
Project Title	The title reflects an issue related to complex engineering problems, showing targets and methods with possible outcomes.	The title reflects an issue related to complex engineering problems, showing targets and methods, but some missing issues.	The title reflects an issue related to the course capstone project, but there may be some missing issues.	The title reflects an issue related to the course capstone project, but it is not complete or specific.	The title does not reflect any issues related to the course capstone project.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						**Total Marks (2)**	

KPIs	Excellent [6-7]	Proficient [4-5]	Good [3]	Acceptable [2]	Unacceptable [1]	No Response [0]	Secured Marks
Survey	The survey developed as a process for complex engineering problems, considering cultural and societal factors, has superior variables, targets, measures, and the implementation process is clear and challenging for future project implementation, with several possible outcomes having good impacts.	The survey, developed as a process for complex engineering problems, considering cultural and societal factors, has good variables, targets, measures, and the implementation process is clear and challenging for future project implementation, with some possible outcomes and little impact.	The survey developed as a process for complex engineering problems, considering cultural and societal factors, has moderate variables, targets, measures, and The implementation process is clear and challenging for future project implementation, with a few possible outcomes and impacts.	The survey, developed as a process for complex engineering problems, considering cultural and societal factors, has good variables, targets, measures, and the implementation process is somewhat clear for future project implementation, with very few possible outcomes and little impact.	The survey developed as a process for complex engineering problems, considering cultural and societal factors, has poor variables, targets, measures, and the implementation process is very unclear for future project implementation, with few possible outcomes but no impacts.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						**Total Marks (7)**	

KPIs	Excellent [4]	Proficient [3]	Good [2]	Acceptable [1]	Unacceptable [0.5]	No Response [0]	Secured Marks
Aims and Objectives	Aims and objectives are written to solve complex engineering problems, considering cultural and societal factors with specific targets, measurements, and implementation processes that are clear and challenging, and have several possible outcomes having very good impacts.	Aims and objectives are written to solve complex engineering problems, considering cultural and societal factors with general targets, measurement, and implementation processes that are not clear and challenging, and have some possible outcomes that have a good impact.	Aims and objectives are written to solve complex engineering problems considering a few cultural and societal factors with narrow targets; measurement, and implementation processes are clear and challenging and have a few possible outcomes, having some impacts.	Aims and objectives are written to solve complex engineering problems considering cultural or societal factors with a very specific target; measurement and implementation processes are not clear or challenging, and have little possible outcome, having no impact.	Aims and objectives are written to solve complex engineering problems, but do not consider cultural and societal factors with any targets; measurement and implementation processes are not clear and challenging, and no possible outcomes that have an impact.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						**Total Marks (4)**	

KPIs	Excellent [5]	Proficient [4]	Good [3]	Acceptable [2]	Unacceptable [1]	No Response [0]	Secured Marks
Literature Review	Specific formats are maintained to review and cite the literature with recent publications. Identified and analyzed the problem correctly.	Specific formats are maintained to review and cite the literature with recent publications. Identified and analyzed the problem correctly, but not all issues were addressed with relevant or intended work.	Specific formats are maintained to review and cite the literature with recent and past publications. Identified and analyzed the problem correctly, but not all issues were addressed with relevant or intended work.	Specific formats are maintained to review and cite the literature with recent and past publications. Identified but could not analyze all the problems correctly, and all issues were not addressed with relevant or intended work.	No specific formats are maintained to review and cite the literature with recent publications. Could not identify and analyze all the problems correctly, and all issues are not addressed with relevant or intended work at all.	No Response at all/ copied from others/ identical submissions with gross errors/ image file printed	
Comments						**Total Marks (5)**	

KPIs	Excellent [5]	Proficient [4]	Good [3]	Acceptable [2]	Unacceptable [1]	No Response [0]	Secured Marks
Experimental Block Diagram	The block diagram is drawn to show the connections of all the possible components or subsystems to show their interdependence with all possible flows of signals from inputs to outputs.	The block diagram is drawn to show the connections of all of the possible components or sub-systems to show their interdependence, with a few missing flows of signals from inputs to outputs.	The block diagram is drawn to show the connections of most of the possible components or sub-systems to show their interdependence, with a few missing flows of signals from inputs to outputs.	The block diagram is drawn to show the connections of a few possible components or sub-systems to show their interdependence, with some missing flow of signals from inputs to outputs.	The block diagram is not drawn to show the connections of all possible components or sub-systems, to show their interdependence and flow of signals from inputs to outputs.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						**Total Marks (5)**	

KPIs	Excellent [4]	Proficient [3]	Good [2]	Acceptable [1]	Unacceptable [0.5]	No Response [0]	Secured Marks
Possible Outcomes	Outcomes are written to achieve complex engineering problems' solutions, considering cultural and societal factors and showing measurement and implementation processes to attain the outcomes with all possible impacts.	Outcomes are written to achieve complex engineering problems' solutions, considering cultural and societal factors and showing measurement and implementation processes to attain the outcomes with some impacts.	Outcomes are written to achieve complex engineering problems' solutions, considering cultural and societal factors, and do not show measurement and implementation processes to attain the outcomes without showing any impacts.	Outcomes are written to achieve complex engineering problems' solutions, but do not consider cultural and societal factors, and do not show measurement and implementation processes to attain the outcomes without showing any impacts.	Outcomes are not written to achieve complex engineering problems' solutions, do not consider cultural and societal factors, and do not show measurement and implementation processes to attain the outcomes without showing any impacts.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						**Total Marks (4)**	

KPIs	Excellent [5]	Proficient [4]	Good [3]	Acceptable [2]	Unacceptable [1]	No Response [0]	Secured Marks
Gantt Chart	Specific formats are maintained to draw the Gantt chart, and there is an order of workflow with all work to be done.	Specific formats are maintained to draw the Gantt chart, and there is an order of workflow with a few tasks missing.	Specific formats are maintained to draw the Gantt chart, and there is an order of workflow with some work missing.	No specific formats are maintained to draw the Gantt chart, and there is little order to the workflow, with some work missing.	No specific formats are maintained to draw the Gantt chart, and there is no order of workflow, with the most important tasks missing.	No Response at all/ copied from others/ identical submissions with gross errors/ image file printed	
Comments						**Total Marks (5)**	

KPIs	Excellent [3]	Proficient [2.5]	Good [2]	Acceptable [1]	Unacceptable [0.5]	No Response [0]	Secured Marks
References	Specific formats are maintained to write references, and all are recently published journal and conference papers having no missing information.	Specific formats are maintained to write the references, and all are journal and conference papers, but some old papers have missing information.	No specific formats are maintained to write the references, and many are internet sources with several missing information and very old references.	No specific formats are maintained to write the references, and most of them are internet sources with missing information.	No specific formats are maintained to write the references, and all are internet sources with missing information.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						**Total Marks (3)**	

An Arduino Based Dual Mode Omni Directional Rover for Intelligent Navigation and Obstacle Avoidance

Introduction

Autonomous mobile robots have become an important area of research in embedded systems due to their ability to perform tasks with minimal human intervention. These systems are widely used in applications such as industrial automation, security, and smart assistance, where efficient navigation and real-time decision-making are essential. This project presents the design and implementation of a **dual-mode omni-directional autonomous rover** using Arduino. The system integrates line following, object tracking, and obstacle avoidance into a single platform, allowing it to operate in both structured and unstructured environments. By utilizing multiple sensors and a multi-mode control strategy, the rover demonstrates intelligent navigation while maintaining a low-cost and practical design suitable for real-world applications.

Survey Summary

Survey Title: User Perception on Multi-Functional Autonomous Robots in Real-World Applications

Number of Respondents: 40

Survey Link: User Perception on Multi-Functional Autonomous Robots in Real-World Applications – Fill out form

Survey Responses: Survey Datasheet (1-40).xlsx

Detailed Summary: Survey Summary.pdf

Purpose

The purpose of this survey is to understand user perception, acceptance, and expectations regarding autonomous robotic systems capable of line following, object tracking, and obstacle avoidance. It focuses on evaluating key factors such as safety, reliability, cost-effectiveness, and real-world usability of such systems. The survey also aims to identify societal and cultural considerations, including user trust, acceptance in daily life, and concerns related to automation.

The survey collected responses from **40 participants**, primarily aged **18–24**, representing a technologically aware group. Their feedback provides valuable insights for designing a system that is both technically effective and socially acceptable.

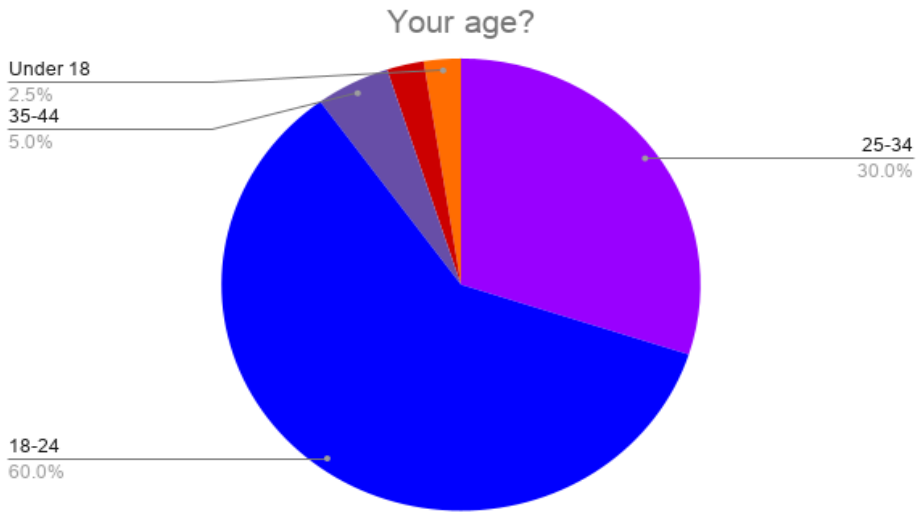


Figure 1: Demographics

Section 1: General Awareness & Acceptance

Majority of respondents demonstrated interest in automated robots:

- 80% are familiar with autonomous robots
- Overall, 65% are comfortable with robots operating in public or personal spaces.

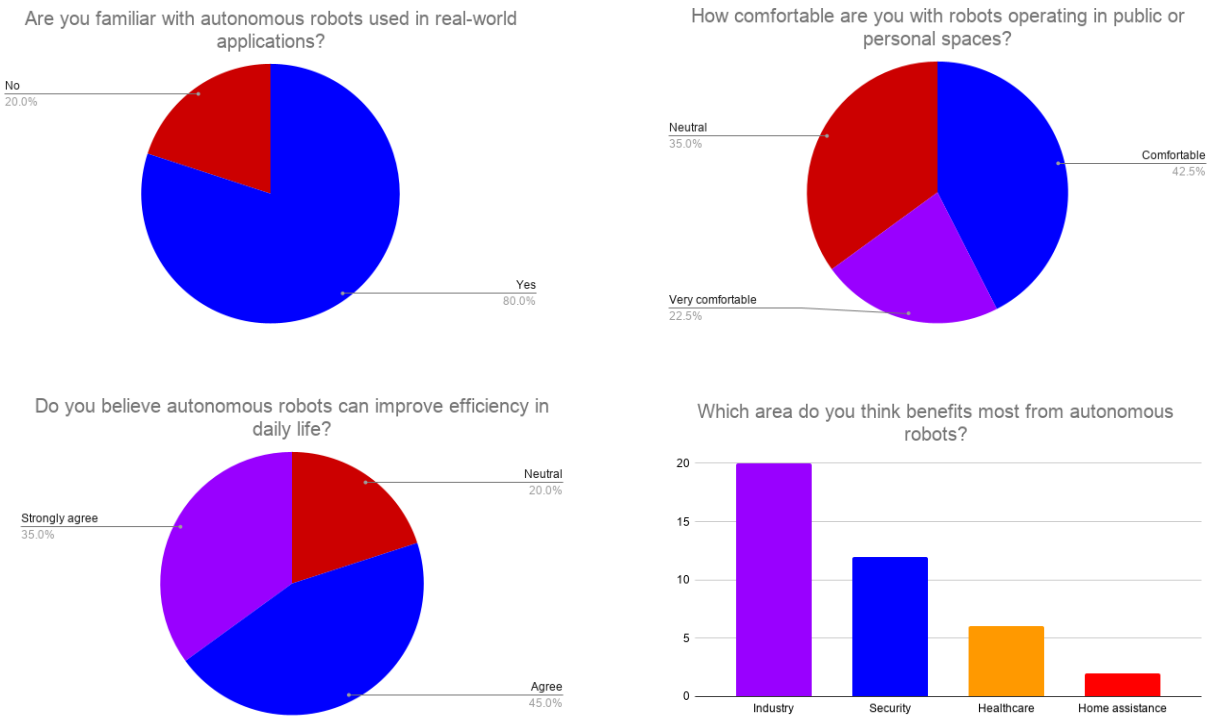


Figure 2: General Interest & Acceptance

Interpretation: The survey indicates a high level of awareness of autonomous robots, with 80% of respondents familiar with them. Around 65% expressed comfort with robots in daily environments, while 35% remained neutral, showing general acceptance but some uncertainty. Additionally, 80% believe robots improve efficiency, reflecting strong confidence in automation. Most respondents (50% industry, 30% security) associate robots with practical, task-oriented applications rather than personal use.

Section 2: Functional Expectations (Engineering Perspective)

- 82.5% believes obstacle avoidance is important.
- 92.5% believes accurate navigation is important.

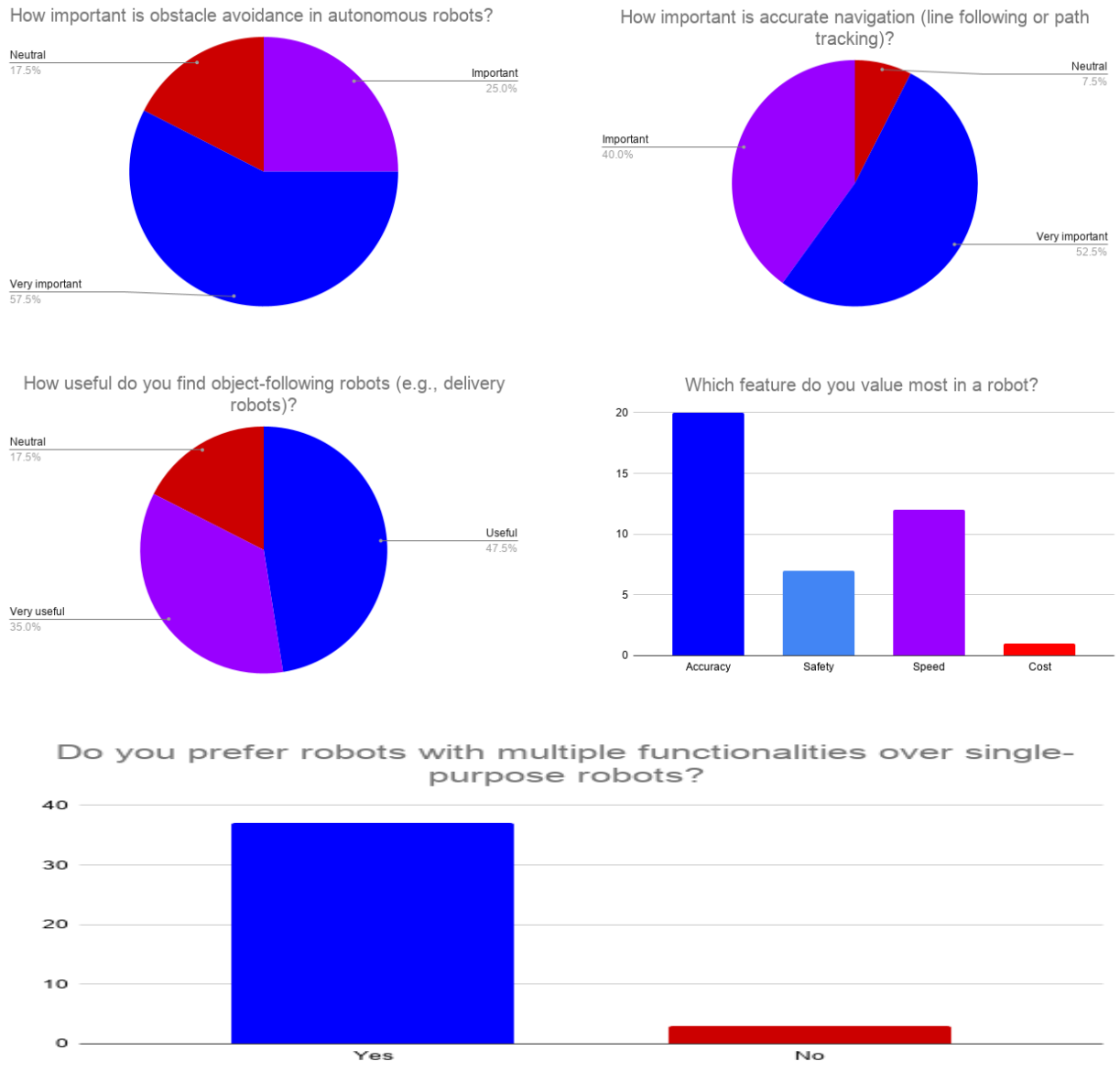


Figure 3: Functional Expectations

Interpretation: The results show that users highly prioritize core engineering features, with **82.5% emphasizing obstacle avoidance** and **92.5% valuing accurate navigation**, highlighting safety and reliability as key requirements. Additionally, **82.5% found object-following useful**, and **50% prioritized accuracy over speed, safety, and cost**. Notably, **92.5% preferred multi-functional robots**, indicating strong demand for versatile and integrated systems.

Section 3: Safety, Reliability & Trust

- 47.5% are concern about system failure
- 95% thinks real-time decision making is important for robots
- Out of 40 participants, 18 thinks human supervision is important

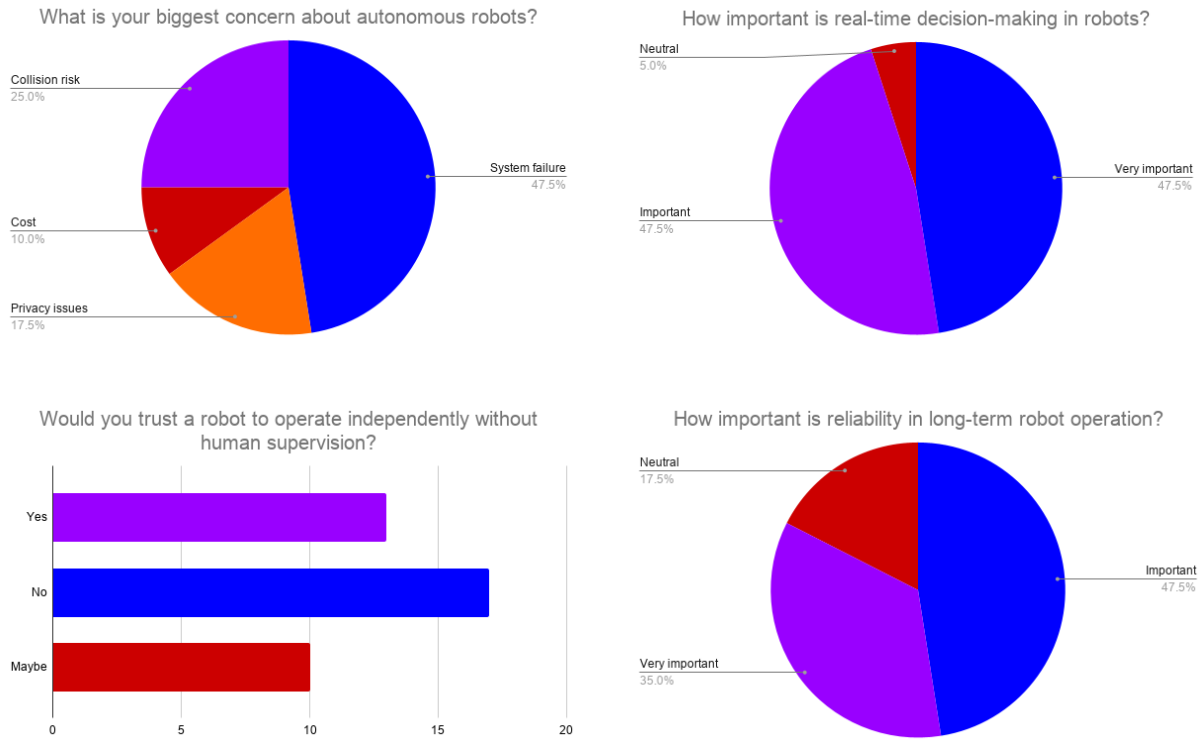


Figure 4: Safety, Reliability & Trust

Interpretation: The findings show that **system failure (47.5%) is the primary concern**, followed by collision risk (25%) and privacy issues (17.5%), highlighting reliability as the main issue. A strong **95% emphasized the importance of real-time decision-making**, yet trust in full autonomy remains limited, with **42.5% lacking trust**. Additionally, **82.5% stressed reliability**, indicating the need for stable and dependable system design.

Section 4: Cultural & Societal Factors

- 22.5% believes autonomous robots won't be accepted in our community.
- 50% of them believe robots will reduce human jobs.

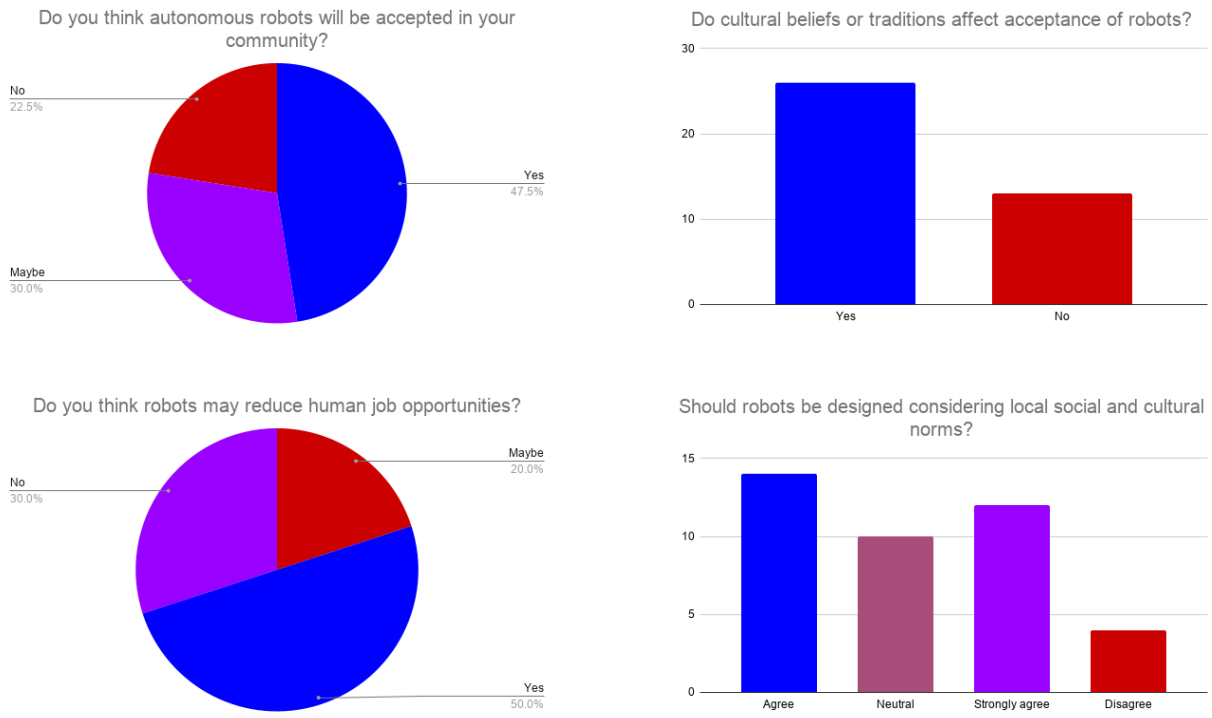


Figure 5: Cultural & Societal Factors

Interpretation: The results show that societal acceptance of robots is still developing, with only **47.5% confident**, while **30% are uncertain** and **22.5% disagree**. Around **65% believe cultural factors influence acceptance**, and **50% are concerned about job loss due to automation**. Additionally, **65% support designing robots considering cultural norms**, highlighting the importance of socially adaptive systems.

Section 5: Real-World Implementation & Engineering Challenges

- 92.5% of people believes cost-effectiveness is important for adopting robotic system in Bangladesh.

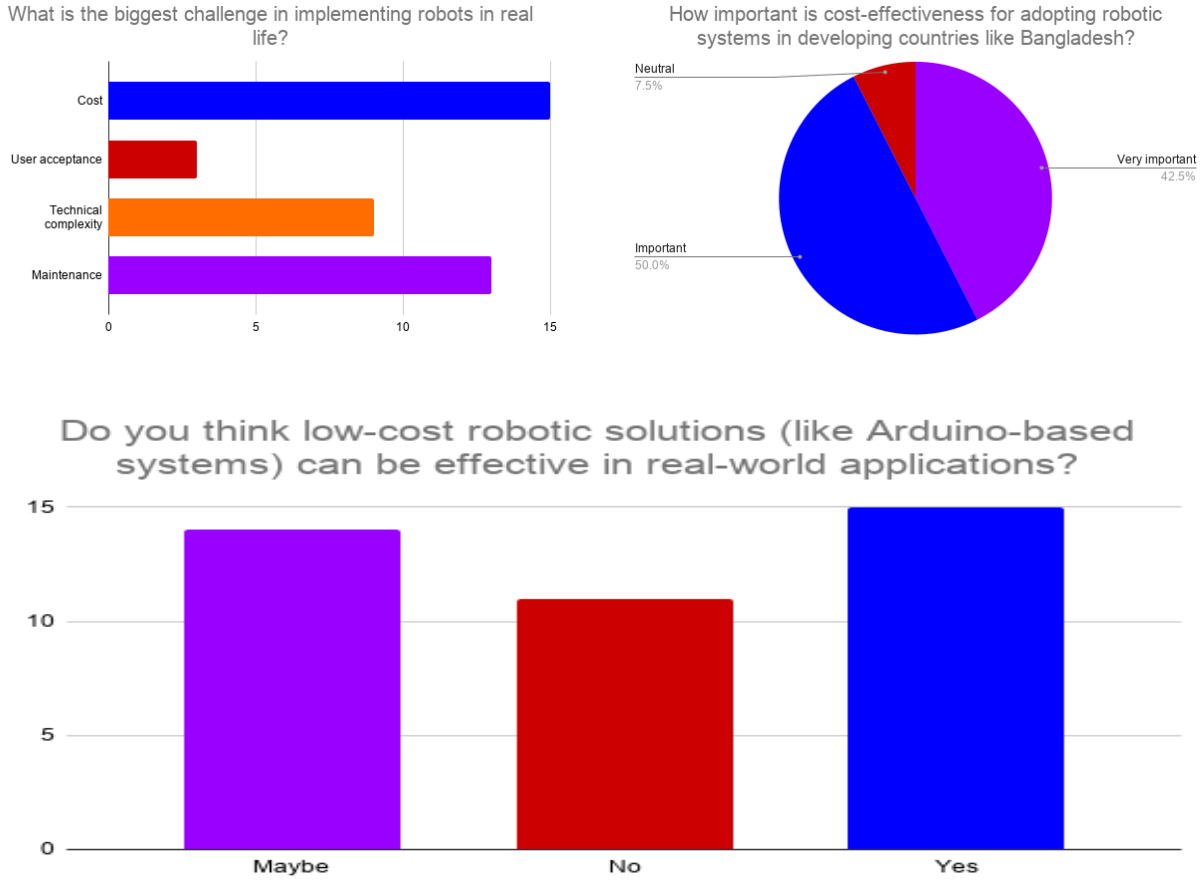


Figure 6: Real-World Implementation & Engineering Challenges

Interpretation: The results show that **cost (37.5%) and maintenance (32.5%) are the main challenges**, followed by technical complexity (22.5%), indicating economic factors are more critical than technical ones. A strong **92.5% emphasized cost-effectiveness**, especially for developing countries. However, opinions on low-cost systems are mixed, with only **37.5% confident in their effectiveness**, while others remain uncertain or skeptical.

Insights: Users demand a low-cost, reliable, and multi-functional robotic system that ensures safety, real-time performance, and social acceptability, which directly supports the design and implementation of the proposed autonomous rover.

Key Comfort Requirements

Open-ended responses and overall survey trends highlight several important comfort requirements for users interacting with autonomous robotic systems:

- **Safe and collision-free operation**, ensuring smooth navigation and reliable obstacle avoidance in dynamic environments.
- **Stable and predictable movement**, avoiding sudden or aggressive motions that may reduce user confidence.
- **High system reliability and real-time responsiveness**, minimizing risks of system failure and delays.
- **Cost-effective implementation**, making the system affordable and practical for real-world use, especially in developing regions.

Conclusion

The survey results indicate strong interest in autonomous robotic systems, particularly those that are **safe, reliable, and multi-functional**. Users emphasized the importance of **accurate navigation, obstacle avoidance, and real-time responsiveness**, while also expressing concerns about **system failure, trust, and job impact**. Additionally, cost-effectiveness and cultural acceptance play a crucial role in real-world adoption. Therefore, the proposed system must balance **technical performance with affordability, safety, and social adaptability** to ensure successful implementation.

Aims and Objectives of the Project

The aim of this project is to design, develop, and evaluate a **dual-mode omni-directional autonomous rover** capable of line following, object tracking, and real-time obstacle avoidance using multi-sensor integration. The system is intended to operate safely, efficiently, and reliably in real-world environments, guided by engineering requirements and user expectations identified through survey analysis.

Objectives

1. Designing the Complete Embedded System Architecture

To design a comprehensive embedded system using Arduino Uno, integrating ultrasonic sensors, IR sensors, servo motor, motor drivers, and omni-directional wheels for autonomous navigation. This objective is supported by survey findings where **80% of respondents believe robots improve efficiency**, indicating strong demand for functional automation systems.

2. Implementing Omni-Directional Movement System

To develop an omni-directional drive mechanism enabling flexible movement (forward, backward, lateral, and diagonal) for improved navigation efficiency. This aligns with user expectations for accurate motion, where **92.5% emphasized the importance of precise navigation**, highlighting the need for advanced movement control.

3. Developing a Multi-Mode Navigation System

To implement a dual-mode system consisting of:

- Line Following Mode
- Object Tracking Mode

This objective is strongly justified by survey results where **92.5% of respondents preferred multi-functional robots**, demonstrating high demand for versatile systems capable of performing multiple tasks.

4. Implementing Real-Time Obstacle Avoidance System

To develop a reliable obstacle detection and avoidance mechanism using IR sensors and ultrasonic sensing to ensure safe navigation. This directly addresses user concerns, as **82.5% identified obstacle avoidance as critical**, and **25% expressed concern about collision risks**, emphasizing safety as a primary requirement.

5. Developing Object Tracking and Line Following Algorithms

To design and implement control algorithms for:

- Line tracking using IR sensors
- Object following using ultrasonic sensor and servo scanning

This is supported by survey findings where **82.5% found object-following useful**, indicating strong acceptance of tracking-based robotic applications.

6. Ensuring Real-Time Decision-Making and System Responsiveness

To implement continuous sensor monitoring and real-time decision-making for adaptive navigation. This objective is based on survey results where **95% of respondents emphasized the importance of real-time responsiveness**, highlighting the need for fast and dynamic system behavior.

7. Enhancing System Reliability and Stability

To ensure stable and consistent system performance through optimized control logic and hardware selection. This is crucial as **47.5% of respondents identified system failure as the primary concern**, and **82.5% emphasized reliability for long-term operation**.

8. Designing a Cost-Effective Robotic System

To develop a low-cost yet efficient robotic solution suitable for practical applications, particularly in developing countries. This objective is supported by survey findings where **92.5% emphasized cost-effectiveness**, and **37.5% identified cost as a major implementation challenge**.

9. Considering Cultural and Societal Factors in Design

To design the system in a way that aligns with societal expectations and cultural acceptance. This is justified by survey results where **65% agreed that cultural factors influence robot acceptance**, and **65% supported designing systems considering social norms**.

10. Evaluating System Performance in Real-World Conditions

To analyze the rover's performance in terms of navigation accuracy, obstacle avoidance efficiency, response time, and overall reliability. These evaluation metrics are aligned with user priorities, where **accuracy (50%) and safety-related features were identified as the most valued aspects**.

In conclusion, the aims and objectives of this project are designed to address complex engineering challenges while incorporating user expectations, cultural considerations, and real-world constraints. Survey insights highlighting the importance of safety, reliability, multi-functionality, and affordability have directly influenced the system design, ensuring that the proposed rover is not only technically effective but also socially acceptable and practically applicable.

Literature Review

Recent research on autonomous mobile robots has focused on low-cost embedded system design for line following, obstacle avoidance, and intelligent navigation. Most systems use microcontroller-based control with infrared and ultrasonic sensing because these components are inexpensive, easy to interface, and suitable for real-time response in educational and prototype robotics. However, the existing literature shows that many designs are still limited to differential-drive motion, single-mode navigation, or controlled indoor environments. More advanced studies improve navigation accuracy and autonomy, but they often require computationally heavy vision, mapping, or learning-based systems that are less suitable for simple Arduino-class embedded platforms.

Shahi et al. [1] developed **LineMaster Pro**, a low-cost intelligent line-following robot with ultrasonic obstacle avoidance. **The system used** an Arduino Nano, dual TCRT5000 IR sensors, an HC-SR04 ultrasonic sensor, an L298N motor driver, two DC motors, and a PID controller with Ziegler–Nichols tuning, along with a finite-state machine for recovery and avoidance. **The results showed** a mean tracking accuracy of **1.18 cm**, **96.7% obstacle detection reliability**, **94% successful recovery from path deviation**, and a **43% improvement** over conventional on-off control, all at a hardware cost of **$28.50**. **However, the system had limitations such as** being designed mainly for educational line-following applications, using a **two-wheel differential drive**, and not addressing dual-mode navigation, object following, or omni-directional mobility.

Bachim et al. [2] developed an autonomous mobile robot capable of both line following and obstacle avoidance for dynamic environments such as warehouses, healthcare facilities, and smart assistance settings. **The system used** an Arduino Uno, **three IR sensors** for line tracking, an **HC-SR04 ultrasonic sensor** for front obstacle sensing, an **L298N motor driver**, DC motors, and an **SG90 servo motor**; the control architecture divided the robot into line-tracking and obstacle-detection subsystems with layered decision logic. **The results showed** that the robot could perform line tracking and obstacle avoidance in structured environments and was considered suitable for practical automation-oriented scenarios. **However, the system had limitations such as** a **short reliable obstacle detection range of about 17.8 cm**, sensitivity to ambient light and surface properties, path deviation at sharp turns, poor adaptability to complex terrain, **no dynamic replanning**, limited battery life, and an **open-loop architecture without encoder or visual feedback**.

Sai Thejas et al. [3] developed a low-cost line follower and obstacle-avoiding robot as a more rigorous engineering study of a tutorial-style platform. **The system used** an Arduino Uno, **two IR reflective sensors**, an **HC-SR04 ultrasonic sensor mounted on an SG90 servo**, an **L298N motor driver**, and a **two-wheel differential drive chassis**. The study also included calibration methodology, test protocols, and analysis under varying lighting and surface conditions. **The results showed** a robust and modular educational robot with clear documentation of sensor physics,

motor control behavior, and environmental effects on performance. **However, the system had limitations such as** remaining a **two-wheel differential-drive platform**, being primarily intended for educational demonstration, and not addressing multi-mode switching, human/object-following behavior, or full 4-wheel omni-directional motion.

Sreedhar et al. [4] developed a prototype line-following and obstacle-detection robot by combining line-following and obstacle-detection modules into one Arduino-based platform. **The system used** an Arduino Uno, **three sets of IR transmitter–receiver sensors**, an **ultrasonic sensor**, a motor shield, DC motors, and a 12 V battery-powered embedded control setup. **The results showed** that the robot could follow a predefined line, detect an obstacle, and take actions to avoid it; the authors also suggested future extensions such as Bluetooth communication, surveillance camera support, image recognition, and off-road capability. **However, the system had limitations such as** focusing on a **basic prototype**, lacking advanced control, lacking quantitative performance evaluation, and not including multi-mode autonomy, sensor fusion for object following, or omni-directional kinematics.

Wenzel et al. [5] developed a vision-based obstacle-avoidance method using **deep reinforcement learning** for autonomous mobile robots. **The system used** a **single monocular camera**, deep convolutional neural networks, predicted depth maps from a generative adversarial network, and DRL algorithms such as DQN and PPO in simulated environments. **The results showed** that **discrete action spaces outperformed continuous control commands** in maze-like environments, and that fusing predicted depth information improved robustness and learning performance. **However, the system had limitations such as** heavy computational requirements, dependence on simulation and camera-based perception, the challenge of transferring from simulation to the real world, and unsuitability for a simple Arduino-class embedded rover due to its high processing demand.

Nguyen and Vu [6] developed an obstacle-avoidance approach for autonomous mobile robots using **ROS**, a **2D costmap**, and the **Dynamic Window Approach (DWA)** for local path planning. **The system used** a laser scanner for environmental sensing, ROS-based navigation architecture, velocity-space optimization, and DWA for generating safe local trajectories in static and dynamic environments. **The results showed** that the robot could generate new paths and work effectively in environments containing both static and dynamic obstacles. **However, the system had limitations such as** reliance on a **laser scanner**, ROS-based computation, and mapping infrastructure, making it more complex and expensive than lightweight microcontroller-based designs. It also addressed obstacle avoidance only, not line following, object tracking, or compact low-cost embedded rover implementation.

Li et al. [7] developed an intelligent tracking and obstacle-avoidance wheel robot based on Arduino. **The system used** Arduino as the core controller, **infrared photoelectric sensors** for

tracking, an **ultrasonic module** for obstacle detection, a motor drive module, and a battery power module. The software was divided into infrared tracking, ultrasonic avoidance, motor drive, and power subsystems. **The results showed** that the robot could perform automatic tracking, automatic obstacle avoidance, automatic searching, and even record travel time, mileage, and speed, demonstrating practical potential for low-cost intelligent wheeled robots. **However, the system had limitations such as** simple **two-sensor line-tracking logic**, traditional ultrasonic avoidance rules, and conventional wheeled chassis without advanced motion control, multi-mode switching, or omni-directional capability.

From the analysis of recent studies, it is evident that most low-cost embedded robotic systems are limited to single or restricted functionalities, primarily focusing on-line following and obstacle avoidance without integrating multiple navigation behaviors such as object tracking and mode switching within a single platform. Additionally, the majority of these systems rely on conventional differential-drive mechanisms, which restrict maneuverability and lack smooth lateral motion, highlighting a significant gap in omni-directional navigation. Furthermore, low-cost sensing techniques using IR and ultrasonic sensors are highly sensitive to environmental conditions such as lighting, surface properties, and limited detection range, reducing reliability in real-world scenarios. Many designs also operate with open-loop or simplistic control strategies, lacking advanced feedback mechanisms, which leads to path deviation and reduced performance in complex environments. On the other hand, advanced robotic systems based on deep learning, vision, or ROS-based path planning provide improved autonomy but require high computational power, expensive hardware, and complex implementation, making them unsuitable for compact, low-cost embedded platforms. Therefore, a clear research gap exists between simple low-cost robots and highly advanced intelligent systems, particularly in achieving a balance between affordability, functionality, and adaptability. The proposed project addresses this gap by introducing a dual-mode omni-directional autonomous rover that integrates line following and object following with real-time obstacle avoidance within a unified Arduino-based embedded system. Unlike existing systems, this design enables mode switching between structured and unstructured navigation while incorporating omni-directional mobility for enhanced maneuverability and control. Thus, the proposed system is unique in providing a cost-effective, multi-functional, and flexible robotic solution that bridges the gap between basic embedded robots and complex autonomous systems, making it highly suitable for practical applications and educational purposes.

Possible Outcome of the Project

The proposed autonomous rover is expected to produce several positive technological and societal outcomes by addressing real-world engineering challenges identified through the survey. Since **80% of respondents believe robots improve efficiency**, the system can enhance automation by performing tasks such as tracking, navigation, and obstacle avoidance with minimal human intervention. With **82.5% emphasizing obstacle avoidance** and **25% concerned about collision risks**, the rover demonstrates how embedded systems can ensure safe and reliable operation in dynamic environments. The implementation of dual-mode functionality directly responds to the **92.5% preference for multi-functional robots**, allowing the system to adapt to different scenarios such as structured path navigation and unstructured object tracking. Furthermore, the integration of real-time decision-making aligns with the **95% of respondents who prioritized responsiveness**, ensuring efficient and adaptive system performance. From an economic perspective, the project promotes affordability and accessibility, as **92.5% of respondents highlighted the importance of cost-effectiveness**, and **37.5% identified cost as a major barrier**. This makes the system suitable for practical deployment in developing countries like Bangladesh. In terms of societal impact, the project contributes to increasing acceptance of robotic systems by addressing concerns such as **system failure (47.5%) and trust issues**, through stable design and predictable behavior. Additionally, by considering that **65% of respondents believe cultural factors influence robot acceptance**, the system encourages the development of socially adaptive and user-friendly technologies.

Key Notes:

- **Improved automation efficiency**, reducing human effort and increasing operational productivity.
- **Enhanced safety and reliability**, addressing user concerns regarding collisions and system failure.
- **Multi-functional robotic capability**, enabling flexible operation through dual-mode navigation.
- **Real-time responsive system behavior**, ensuring adaptive decision-making in dynamic environments.
- **Cost-effective solution**, suitable for widespread adoption in resource-limited settings.
- **Increased societal acceptance of robotics**, by aligning system design with user expectations and cultural considerations.
- **Educational and technological impact**, encouraging innovation and learning in embedded systems and robotics.

Experimental Block Diagram

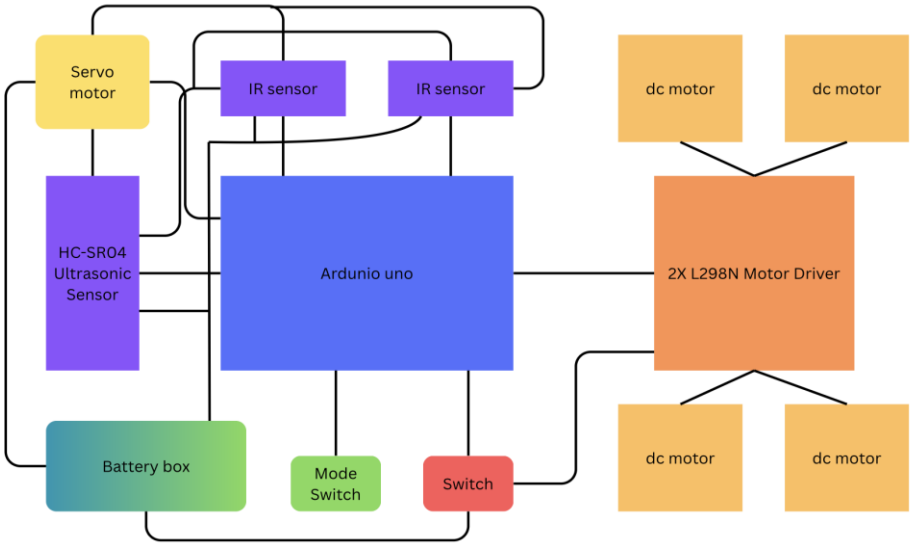


Figure 7: Proposed Block Diagram

The block diagram represents the overall architecture of the proposed dual-mode autonomous rover system, illustrating the interaction between sensing, processing, control, and actuation units. At the core of the system is the **Arduino Uno**, which acts as the main embedded controller. It receives input signals from multiple sensors and processes them to generate appropriate control signals for the rover's movement. The system uses two **IR sensors**, which are connected to the Arduino for **line detection and obstacle sensing**. These sensors enable the rover to follow a predefined path in line-following mode and assist in detecting nearby obstacles. An **HC-SR04 ultrasonic sensor**, mounted on a **servo motor**, is used for distance measurement and environmental scanning. The servo motor allows the ultrasonic sensor to rotate, enabling the system to detect objects from different directions. This setup is primarily used in **object-following mode** to identify and track the nearest object. A **mode switch** is connected to the Arduino to select between the two operational modes: line following and object following. Additionally, a **power switch** controls the overall power supply of the system. The processed signals from the Arduino are sent to **two L298N motor driver modules**, which provide sufficient current to drive the **four DC motors** connected to the omni wheels. These motor drivers enable independent control of each motor, allowing the rover to achieve **omni-directional movement**, including forward, backward, lateral (sideways), diagonal, and rotational motion. The entire system is powered by a **battery box**, which supplies energy to the Arduino, sensors, and motor drivers. Overall, the block diagram demonstrates a complete embedded system where sensor inputs, control logic, and motor actuation are integrated to achieve **dual-mode navigation, omni-directional mobility, and real-time obstacle avoidance**.

PROJECT TIMELINE (7 Weeks)

Start: 15th March, 2026

Projected Deadline: 2nd May, 2026

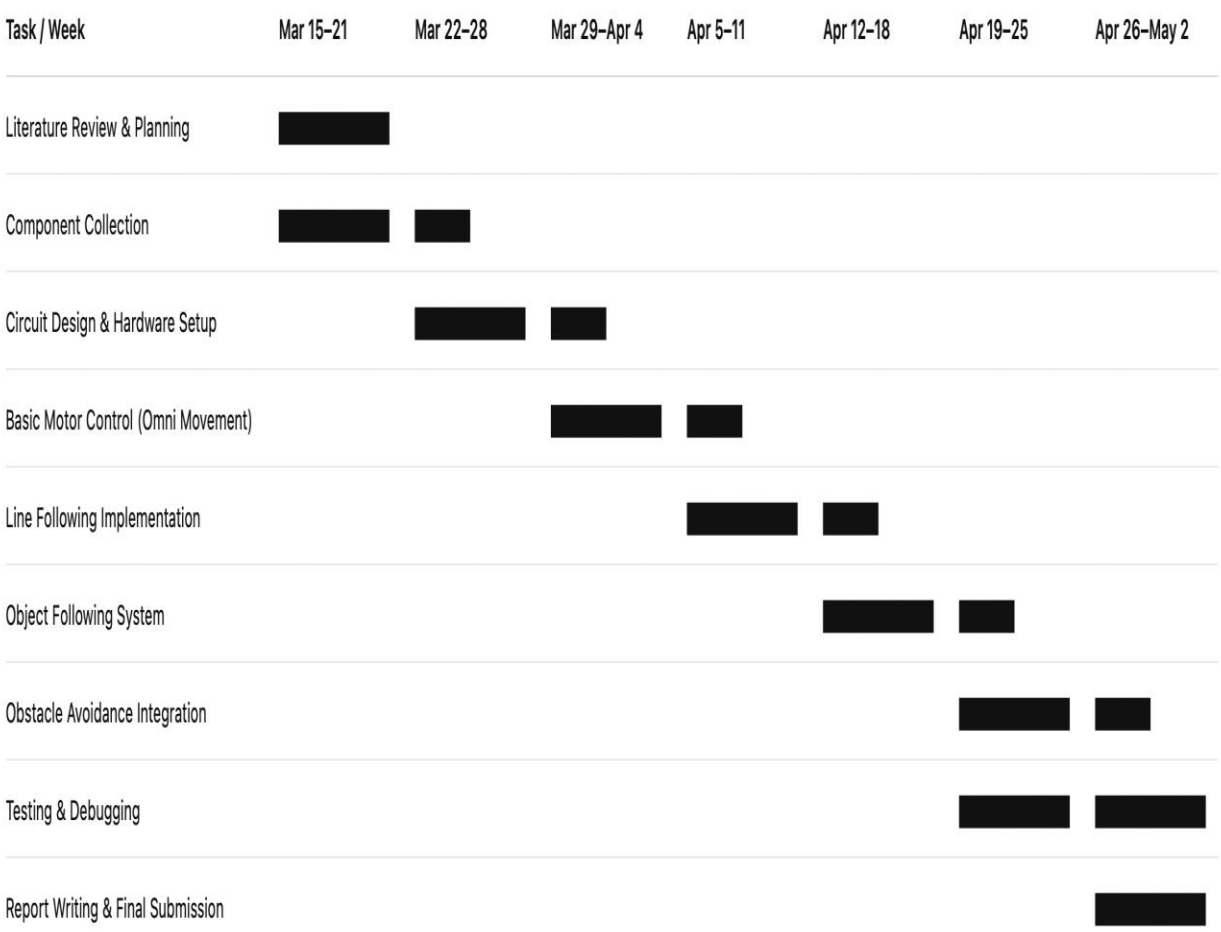


Figure 7: Gantt Chart

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